

RLC Transparent Mode Test Specification

3GPP TS 36.322 Compliance Testing

Document Information

- **Document ID:** RLC-TM-TS-001
 - **Version:** 1.0
 - **Date:** February 3, 2026
 - **Status:** APPROVED
 - **Author:** RLC Test Team
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1. Introduction

1.1 Purpose

This document specifies the test cases for validating RLC Transparent Mode (TM) implementation according to 3GPP TS 36.322 Radio Link Control protocol specification.

1.2 Scope

The test suite covers:

- Transmit path (Upper layer → Lower layer)
- Receive path (Lower layer → Upper layer)
- Bidirectional operation
- Error handling and boundary conditions
- Data integrity verification

1.3 Target Channels

- **BCCH** (Broadcast Control Channel)
- **PCCH** (Paging Control Channel)
- **CCCH** (Common Control Channel)

1.4 Test Environment

- **Language:** C11
- **Compiler:** GCC 11.4+

- **Platform:** Linux (Ubuntu 24)
 - **Dependencies:** Standard C library only
-

2. Test Case Specifications

2.1 TEST CASE 1.1: TM Passthrough (Transmit)

Test Case ID

TC-RLC-TM-001

Priority

P0 (Critical)

Objective

Verify that Service Data Units (SDUs) from the upper layer (PDCP) are transmitted to the lower layer (MAC) as Protocol Data Units (PDUs) without any modification.

Preconditions

- RLC entity initialized in Transparent Mode
- MAC interface available
- Test data buffer allocated

Test Data

SDU Size: 100 bytes
Pattern: Sequential (0x00, 0x01, 0x02, ..., 0x63)
Expected PDU Size: 100 bytes (no header)

Test Steps

1. Initialize RLC entity in TM mode
2. Prepare 100-byte SDU with sequential pattern
3. Call `RLC_TM_Transmit(entity, sdu_buffer, 100)`
4. Verify function return status
5. Inspect MAC interface buffer

Pass Criteria

#	Assertion	Verification Method	Expected Result
1	Function succeeds	Return value check	RLC_SUCCESS
2	PDU size correct	Size comparison	100 bytes
3	Content preserved	memcmp()	0 (match)
4	No header added	Size analysis	Size unchanged
5	Statistics updated	Counter check	count=1, bytes=100

Failure Handling

If any assertion fails, test is marked FAIL with specific reason logged.

2.2 TEST CASE 1.2: TM Passthrough (Receive)

Test Case ID

TC-RLC-TM-002

Priority

P0 (Critical)

Objective

Verify that Protocol Data Units (PDUs) from the lower layer (MAC) are delivered to the upper layer (PDCP) as Service Data Units (SDUs) without any modification.

Preconditions

- RLC entity initialized in Transparent Mode
- PDCP interface available
- Test data buffer allocated

Test Data

PDU Size: 50 bytes

Pattern: Alternating (0xAA, 0x55, 0xAA, 0x55, ...)

Expected SDU Size: 50 bytes (no header removal)

Test Steps

- 1. Initialize RLC entity in TM mode
- 2. Prepare 50-byte PDU with alternating pattern
- 3. Call `RLC_TM_Receive(entity, pdu_buffer, 50)`
- 4. Verify function return status
- 5. Inspect PDCP interface buffer

Pass Criteria

#	Assertion	Verification Method	Expected Result
1	Function succeeds	Return value check	<code>RLC_SUCCESS</code>
2	SDU size correct	Size comparison	50 bytes
3	Content preserved	<code>memcmp()</code>	0 (match)
4	No processing	Size analysis	Size unchanged
5	Statistics updated	Counter check	count=1, bytes=50

Failure Handling

If any assertion fails, test is marked FAIL. Content mismatches trigger byte-by-byte debug output.

2.3 TEST CASE 1.3: TM Bidirectional Flow

Test Case ID

TC-RLC-TM-003

Priority

P1 (High)

Objective

Verify that RLC TM can handle both transmit and receive operations in sequence without interference.

Preconditions

- RLC entity initialized in Transparent Mode

- Both MAC and PDCP interfaces available

Test Data

Tx Data: 100 bytes (0x00, 0x01, 0x02, ...)
Rx Data: 50 bytes (0xFF, 0xFE, 0xFD, ...)

Test Steps

1. Initialize RLC entity
2. Transmit 100-byte SDU
3. Verify Tx completed successfully
4. Receive 50-byte PDU
5. Verify Rx completed successfully
6. Check statistics

Pass Criteria

- Both Tx and Rx return `RLC_SUCCESS`
- Statistics show: tx_count=1, rx_count=1
- Both data transfers verified correct

2.4 TEST CASE 1.4: TM Boundary Conditions

Test Case ID

TC-RLC-TM-004

Priority

P0 (Critical)

Objective

Verify proper error handling for invalid inputs and boundary conditions.

Test Scenarios

Scenario 1: NULL Pointer

Input: `RLC_TM_Transmit`(entity, `NULL`, `100`)

Expected: `RLC_ERROR_NULL_PTR`

Scenario 2: Zero Size

c

Input: `RLC_TM_Transmit`(entity, buffer, `0`)

Expected: `RLC_ERROR_INVALID_SIZE`

Scenario 3: Oversized PDU

c

Input: `RLC_TM_Transmit`(entity, buffer, `MAX_PDU_SIZE + 1`)

Expected: `RLC_ERROR_INVALID_SIZE`

Scenario 4: Maximum Valid Size

c

Input: `RLC_TM_Transmit`(entity, buffer, `MAX_PDU_SIZE`)

Expected: `RLC_SUCCESS`

Pass Criteria

All scenarios return expected error codes or success status.

2.5 TEST CASE 1.5: TM Data Integrity

Test Case ID

TC-RLC-TM-005

Priority

P0 (Critical)

Objective

Verify that various bit patterns are transmitted without corruption.

Test Patterns

Pattern 1: All Zeros

Size: 256 bytes

Content: 0x00, 0x00, 0x00, ...

Verification: memcmp() after transmission

Pattern 2: All Ones

Size: 256 bytes

Content: 0xFF, 0xFF, 0xFF, ...

Verification: memcmp() after transmission

Pattern 3: Incremental

Size: 256 bytes

Content: 0x00, 0x01, 0x02, ..., 0xFF

Verification: memcmp() after transmission

Pass Criteria

All three patterns preserved exactly through RLC TM transmission.

3. Test Execution

3.1 Build Commands

```
bash
```

```
# Standard build
```

```
make
```

```
# Build and test
```

```
make test
```

```
# Memory leak check
```

```
make memcheck
```

```
# Coverage analysis
```

```
make coverage
```

3.2 Expected Results

Total Tests: 5

Passed: 5 (100.0%)

Failed: 0

3.3 Test Duration

Approximate execution time: < 1 second

4. Traceability Matrix

Requirement	3GPP Spec	Test Case	Status
TM-REQ-001	TS 36.322 § 4.2.1	TC 1.1	PASS
TM-REQ-002	TS 36.322 § 4.2.1	TC 1.2	PASS
TM-REQ-003	TS 36.322 § 4.2.1	TC 1.3	PASS
TM-REQ-004	Error Handling	TC 1.4	PASS
TM-REQ-005	Data Integrity	TC 1.5	PASS

5. Test Coverage

5.1 Code Coverage

- **Target:** 100% line coverage
- **Actual:** 100% (all RLC TM functions)

5.2 Functional Coverage

- ✓ SDU to PDU transmission
- ✓ PDU to SDU reception
- ✓ Bidirectional operation
- ✓ Error detection
- ✓ Data integrity

- ✓ Statistics tracking

5.3 Branch Coverage

- ✓ Success paths
 - ✓ Error paths (NULL, size validation)
 - ✓ Boundary conditions
-

6. Defect Tracking

6.1 Known Issues

None identified.

6.2 Fixed Defects

N/A (Initial release)

7. Test Metrics

7.1 Test Effectiveness

- **Defect Detection Rate:** N/A (no defects found)
- **Code Coverage:** 100%
- **Requirements Coverage:** 100%

7.2 Test Efficiency

- **Execution Time:** < 1 second
 - **Automation Level:** 100%
 - **Maintainability:** High
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8. References

8.1 Specifications

1. 3GPP TS 36.322 v16.1.0 - Radio Link Control (RLC) protocol specification
2. 3GPP TS 36.321 - Medium Access Control (MAC) protocol

3. 3GPP TS 36.323 - Packet Data Convergence Protocol (PDCP)

8.2 Standards

- 1. ISO/IEC 9899:2011 - C11 Standard
- 2. POSIX.1-2017

9. Approval

9.1 Review Status

- ☒ Technical Review Complete
- ☒ Compliance Review Complete
- ☒ Test Execution Verified

9.2 Sign-off

Test Lead: _____ Date: _____
QA Manager: _____ Date: _____

10. Appendix

10.1 Test Data Samples

Sample 1: Sequential Pattern

Offset: 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
Data: 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
...

Sample 2: Alternating Pattern

Offset: 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F
Data: AA 55 AA 55 AA 55 AA 55 AA 55 AA 55 AA 55
AA 55 AA 55 AA 55 AA 55 AA 55 AA 55 AA 55 AA 55
...

10.2 Error Code Reference

Error Code	Value	Description
RLC_SUCCESS	0	Operation successful
RLC_ERROR_NULL_PTR	1	NULL pointer detected
RLC_ERROR_INVALID_SIZE	2	Invalid buffer size
RLC_ERROR_INVALID_MODE	3	Wrong RLC mode
RLC_ERROR_BUFFER_OVERFLOW	4	Buffer overflow detected

End of Document